HCF and LCM: answer key

1 a) $24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$ · b) $45 = 3 \times 3 \times 5 = 3^2 \times 5$ · c) $84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$

2 a) $72 = 2 \times 2 \times 2 \times 3 \times 3 = 2^3 \times 3^2$ · b) $90 = 2 \times 3 \times 3 \times 5 = 2 \times 3^2 \times 5$

3 a) $18 = 2 \times 3 \times 3$, $27 = 3 \times 3 \times 3$: HCF 9, LCM 54
b) $20 = 2 \times 2 \times 5$, $30 = 2 \times 3 \times 5$: HCF 10, LCM 60
c) $28 = 2 \times 2 \times 7$, $42 = 2 \times 3 \times 7$: HCF 14, LCM 84
d) $8 = 2 \times 2 \times 2$, $9 = 3 \times 3$: no shared primes, HCF 1, LCM 72

4 24 is the LCM (the smallest number in both times tables). The HCF of 8 and 12 is 4.

5 a) HCF of 24 and 36 = 12, so 12 bracelets, each with 2 red and 3 blue beads.
b) LCM of 12 and 18 = 36, so after 36 seconds.

6 a) HCF 12, so 3/5 · b) HCF 14, so 2/3

? 1) 7 and 5 are both prime, so they share nothing: HCF 1 and LCM $7 \times 5 = 35$. 2) The LCM is a multiple of each number, so it is at least as big as the bigger one; it is bigger than both unless one number divides the other.

★ **Challenge:** $4 \times 24 = 96 = 8 \times$ (other), so the other number is 12. Check: HCF(8, 12) = 4 and LCM(8, 12) = 24.